A literature comparison matrix

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Ask which fields align, differ or remain unknown.

Teach evidence organization without silently treating missing information as agreement.

Keep one habit: Model task/metric mismatch → preserve raw denominator gaps → compare equal-rate conditions → assess recall/precision mismatch → specify common evaluation.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of a literature comparison matrix

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Replication and reproduction. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of replication and reproduction

Reference example, not the only acceptable wording: Teacher explanation: Reproducing a computation uses recorded artifacts; replication collects new **evidence** under a comparable question.

Naming conventions vary, so define the terms in the project. Here replay means same artifacts/procedure; new-data replication means newly collected eligible cases under the same plan.

Preserve environment, data versions and raw results.

Without opening the old lesson, explain one key idea from Finding primary research. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of finding primary research

Reference example, not the only acceptable wording: Teacher explanation: Primary research presents a study's own methods and results. A summary can help discovery but should not replace inspection of the original **evidence**.

Record title, authors, version, date, identifier and access location. Distinguish paper, code, data and commentary. Verify that a quoted claim belongs to the cited version.

Pictured end goal

Explain a literature comparison matrix using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Replication and reproduction

Finding primary research

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. A **literature matrix** compares studies using aligned columns rather than sorting headline numbers with different meanings. Use the supplied constructed scenarios to make the **mechanism** inspectable.

Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of a literature comparison matrix

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach matrix fields as evidence for comparability judgments

> Keep known differences, missing values and numeric results distinct.

Definitions and complete fictional matrix contract

A **literature matrix** compares studies using aligned columns rather than sorting headline numbers with different meanings.

Record question, data, unit, baseline, metric, split, resources and limitations. Mark incomparable fields rather than forcing a ranking.

Matrix rule	Complete fictional comparison protocol
Study identity	Keep exact source/version/locator when known; supplied examples have no real citation
Question and task	Name what each study evaluates before comparing outcomes
Cases and unit	Record dataset/population, sample denominator and experimental unit; mark absent details not reported
Baseline and metric	Keep reference method, outcome definition, cutoff and aggregation rule
Split and resources	Record training/evaluation separation, environment and resource measurements if supplied
Limitations	State known task/condition gaps and unknown methods
Comparison judgment	Aligned for a stated purpose / different known fields / insufficiently reported; no automatic rank
No invented counts	A percentage alone does not supply its raw numerator or denominator
Scope	Fictional matrix practice, not real literature synthesis, pooled meta-analysis or tool recommendation

Main filled matrix and checked reasoning

Main matrix field	Study A supplied record	Study B supplied record
Task	Classification, described as easy	Generated answers, described as difficult
Reported metric/result	Accuracy90%	Claim support 80%
Dataset/corpus and version	Not reported	Not reported
Evaluation unit	Classification cases implied; exact definition not reported	Claim versus whole-answer aggregation not reported
Raw count/denominator	Not reported; do not invent 9/10	Not reported; do not invent 8/10
Baseline/model version	Not reported	Not reported
Split and resource conditions	Not reported	Not reported
Known comparison gap	Different task/outcome from B	Different task/outcome from A
Valid conclusion	A's headline percentage is numerically higher	No direct cross-task superiority follows

Main classification accuracy 90% and generated-answer claim support 80% differ in task/metric; denominators, versions, baseline, split and resources not reported. Do not invent 9/10 or8/10. Picture S1 indoor 8/10 and S2 outdoor 16/20 both 80%, but neither same-photo controlled comparison nor equivalence follows. Removing condition hides known context. Two unknown fields are not confirmed matches.

Selected recall@5 .8 versus precision@3 .9 on another corpus cannot be directly ranked: metric, cutoff and corpus differ. Retain aggregation details as unknown rather than inventing per-query counts; align intended task and judged data before a meaningful comparison.

A literature comparison matrix — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Study A reports accuracy 90% on easy classification; B reports claim support 80% on difficult generated answers.	Record question, data, unit, baseline, metric, split, resources and limitations. Mark incomparable fields rather than forcing a ranking.	Different tasks and metrics cannot establish A superior to B. Keep separate columns and identify a common evaluation needed.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Different tasks and metrics cannot establish A superior to B. Keep separate columns and identify a common evaluation needed.

Qualification: A matrix documents comparison limits; it does not create missing common **evidence**.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Study A reports accuracy 90% on easy classification; B reports claim support **80%** on difficult generated answers.

Different tasks and metrics cannot establish A superior to B. Keep separate columns and identify a common evaluation needed.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Missing-field, equivalence and pooling probes

Literature-matrix probe	Expected interpretation
90% and 80% use different tasks/metrics	Do not rank tools by the scalar numbers alone
Same80% on different samples	Equal observed rates, not established equivalence
Denominator absent	Mark not reported; no reconstructed raw count
Indoor/outdoor field removed	Important known condition hidden
Both sources omit a field	Unknown agreement, not proven matching conditions
Different metric/cutoff/corpus	No direct score ranking; align the intended comparison first
One study has more cases	More reported cases alone does not solve task or quality mismatch
Average incompatible percentages	No meaningful pooled result without a justified common estimand/protocol
Matrix completed	Evidence organization, not new experiment or causal synthesis

Separate indoor/outdoor comparison and intact denominators

A **literature matrix** that preserves differences. Two fictional studies evaluate related tools under different conditions.

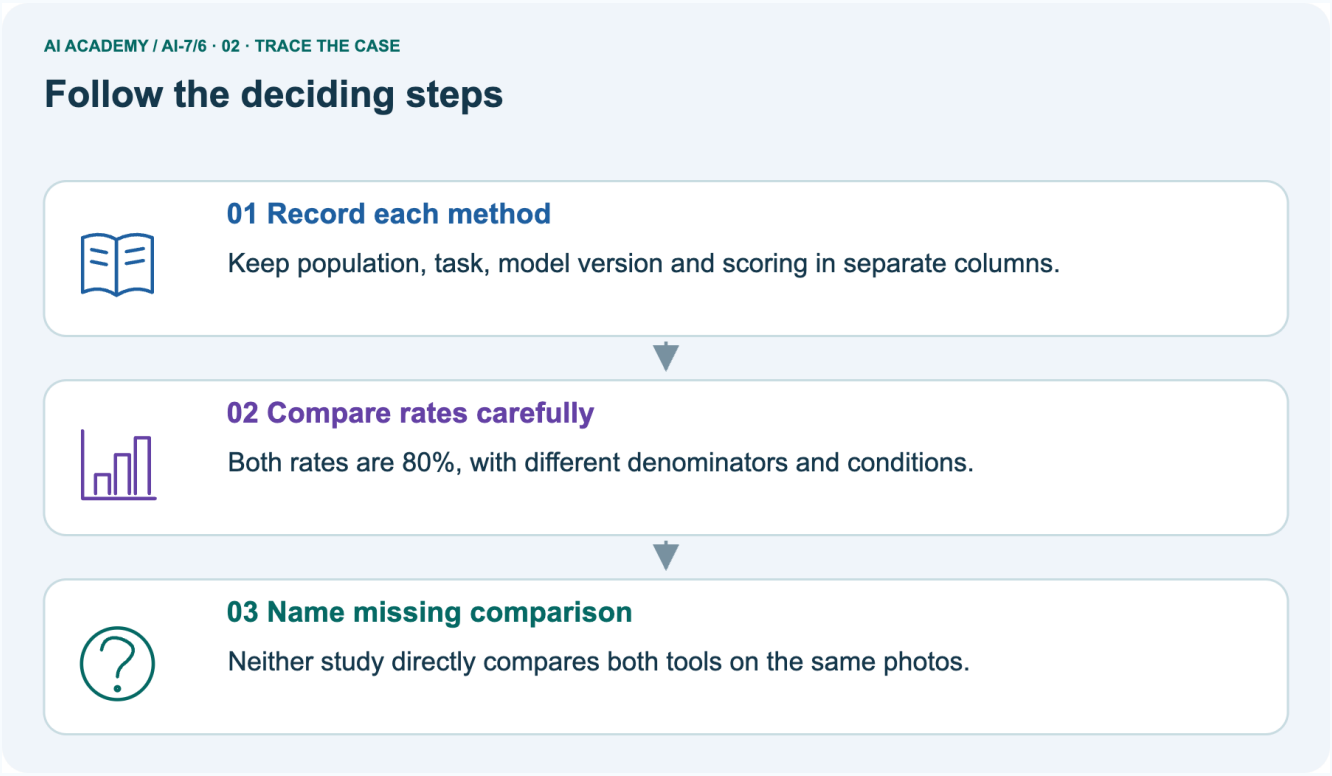
Study S1: 8/10 correct on indoor photos. S2: 16/20 on outdoor photos. Both report **80%**.

The picture is a separate photo-study comparison. S1 reports 8/10 correct on indoor photos; S2 reports 16/20 on outdoor photos. Both rates are 80%, but the samples, denominators and conditions differ. Neither supplied record directly evaluates both tools on the same photos. Equal percentages do not establish equivalent performance or identical uncertainty.

Removing the indoor/outdoor column makes the matrix less informative while leaving the original difference intact. To answer a shared application question, first define an appropriate common task, eligible data, scoring, versions and conditions; a matrix by itself does not run that evaluation.

Separate picture matrix field	S1	S2
Photo condition	Indoor	Outdoor
Correct / evaluated	8/10	16/20
Observed correctness rate	80%	80%
Exact photo IDs/population	Not reported	Not reported
Tool/model version and baseline	Not reported	Not reported
Scoring comparability	Correctness reported; exact label protocol not supplied	Correctness reported; exact label protocol not supplied
Split/resources	Not reported	Not reported
Direct same-photo A/B evaluation	Not supplied	Not supplied
Interpretation	Indoor supplied-set result	Outdoor supplied-set result; no direct equivalence/ranking established

Picture trace and removed-condition key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Record each method	Keep population, task, model version and scoring in separate columns.
2. Compare rates carefully	Both rates are 80%, with different denominators and conditions.
3. Name missing comparison	Neither study directly compares both tools on the same photos.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **literature matrix** helps compare **evidence** without flattening differences. Equal percentages do not mean equivalent tasks, samples or methods.

These fictional studies illustrate comparison practice; no real literature finding is asserted.

Change: Remove the indoor/outdoor column.

Prediction: The matrix conceals an important condition difference.

Reason: A reader might now mistake unlike studies for a controlled head-to-head comparison.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Literature matrix	A structured comparison of study methods and results
Comparable metric	A measure with aligned definition and scope
Evidence gap	Information needed but not supplied

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: A literature comparison matrix

The facts we will use

AI ACADEMY / AI-7/6

Same rate, different study conditions

	Correct	Total	Condition
Study S1	8	10	Indoor
Study S2	16	20	Outdoor

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/6

A literature comparison matrix

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Record each method

Keep population, task, model version and scoring in separate columns.



2. Compare rates carefully

Both rates are 80%, with different denominators and conditions.



3. Name missing comparison

Neither study directly compares both tools on the same photos.

CHANGE ONE THING

Remove the indoor/outdoor column.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Study A reports accuracy 90% on easy classification; B reports claim support **80%** on difficult generated answers. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Different tasks and metrics cannot establish A superior to B. Keep separate columns and identify a common evaluation needed.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for A literature comparison matrix. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A matrix documents comparison limits; it does not create missing common **evidence**.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the **literature matrix** preserve?

Keep population, task, model version and scoring in separate columns. Both rates are **80%**, with different denominators and conditions. Neither study directly compares both tools on the same photos.

Changed case: The matrix conceals an important condition difference. A reader might now mistake unlike studies for a controlled head-to-head comparison.

Changed-case reasoning: The matrix conceals an important condition difference. A reader might now mistake unlike studies for a controlled head-to-head comparison.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main 90/80% numbers cannot rank classification versus generated-answer support; unknown fields stay not reported. Picture both 80% but different indoor/outdoor cases and denominators do not establish equivalence or direct ranking. Selected recall@5 .8 versus precision@3 .9 on another corpus has different metric, cutoff and data; align task and judged data before comparison.

Do not invent aggregation or raw counts from the decimals.

Diagnose the current misconceptions

The worked result establishes every future case: A matrix documents comparison limits; it does not create missing common **evidence**.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: A has recall@5 .8; B has precision@3 .9 on another corpus. Can you rank them?

Teacher explanation: No directly comparable ranking: cutoff, metric and corpus differ. Align task and judged data before comparison.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: A **literature matrix** compares studies using aligned columns rather than sorting headline numbers with different meanings. Record question, data, unit, baseline, metric, split, resources and limitations.

Mark incomparable fields rather than forcing a ranking.

W2 Trace the worked case

Study A reports accuracy 90% on easy classification; B reports claim support **80%** on difficult generated answers. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Different tasks and metrics cannot establish A superior to B. Keep separate columns and identify a common evaluation needed.

W3 Audit the limit

A report omits this qualification: A matrix documents comparison limits; it does not create missing common **evidence**. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A matrix documents comparison limits; it does not create missing common **evidence**.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A matrix documents comparison limits; it does not create missing common **evidence**.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Different tasks and metrics cannot establish A superior to B. Keep separate columns and identify a common evaluation needed.

Qualification: A matrix documents comparison limits; it does not create missing common **evidence**.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

A has recall@5 .8; B has precision@3 .9 on another corpus. Can you rank them?

Teacher explanation: No directly comparable ranking: cutoff, metric and corpus differ. Align task and judged data before comparison.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: No directly comparable ranking: cutoff, metric and corpus differ. Align task and judged data before comparison. Require intermediate reasoning and the stated scope. Fresh retry: One study reports mean latency, another median on different hardware. Expected reasoning: Record statistic/workload/hardware differences; headline values are not a matched comparison. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled a literature comparison matrix **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of a literature comparison matrix.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Hypotheses and measurable outcomes. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/7, ai-7/9, ai-7/12. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of a literature comparison matrix

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Study A reports accuracy 90% on easy classification; B reports claim support **80%** on difficult generated answers. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Different tasks and metrics cannot establish A superior to B. Keep separate columns and identify a common evaluation needed.

Check every learner before discussion: Use the worked case for A literature comparison matrix. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A matrix documents comparison limits; it does not create missing common **evidence**.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Remove the indoor/outdoor column. Require a **prediction** and a reason before comparing. Expected result: The matrix conceals an important condition difference.

Reason: A reader might now mistake unlike studies for a controlled head-to-head comparison.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

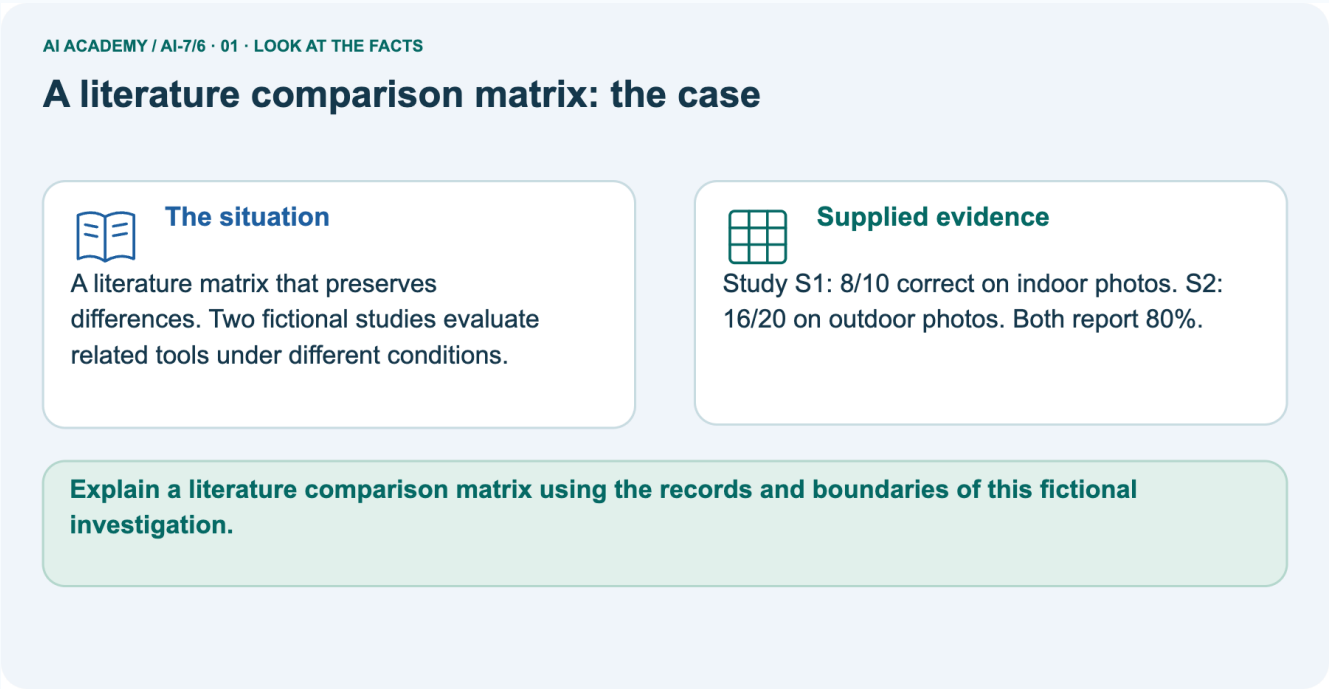
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. A literature comparison matrix: the case



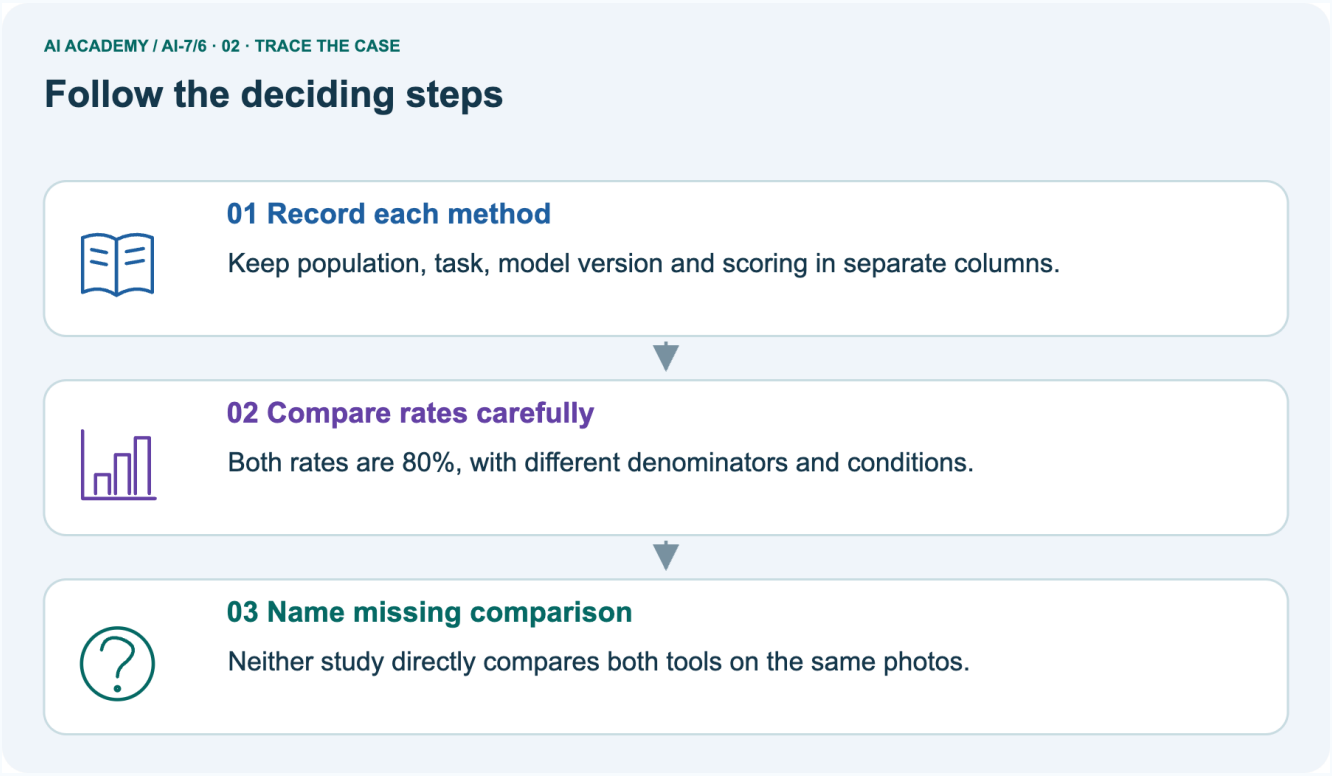
Follow the labelled picture with the matching explanation. Open the vector diagram

A **literature matrix** that preserves differences. Two fictional studies evaluate related tools under different conditions. Supplied facts: Study S1: 8/10 correct on indoor photos. S2: 16/20 on outdoor photos. Both report **80%**.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



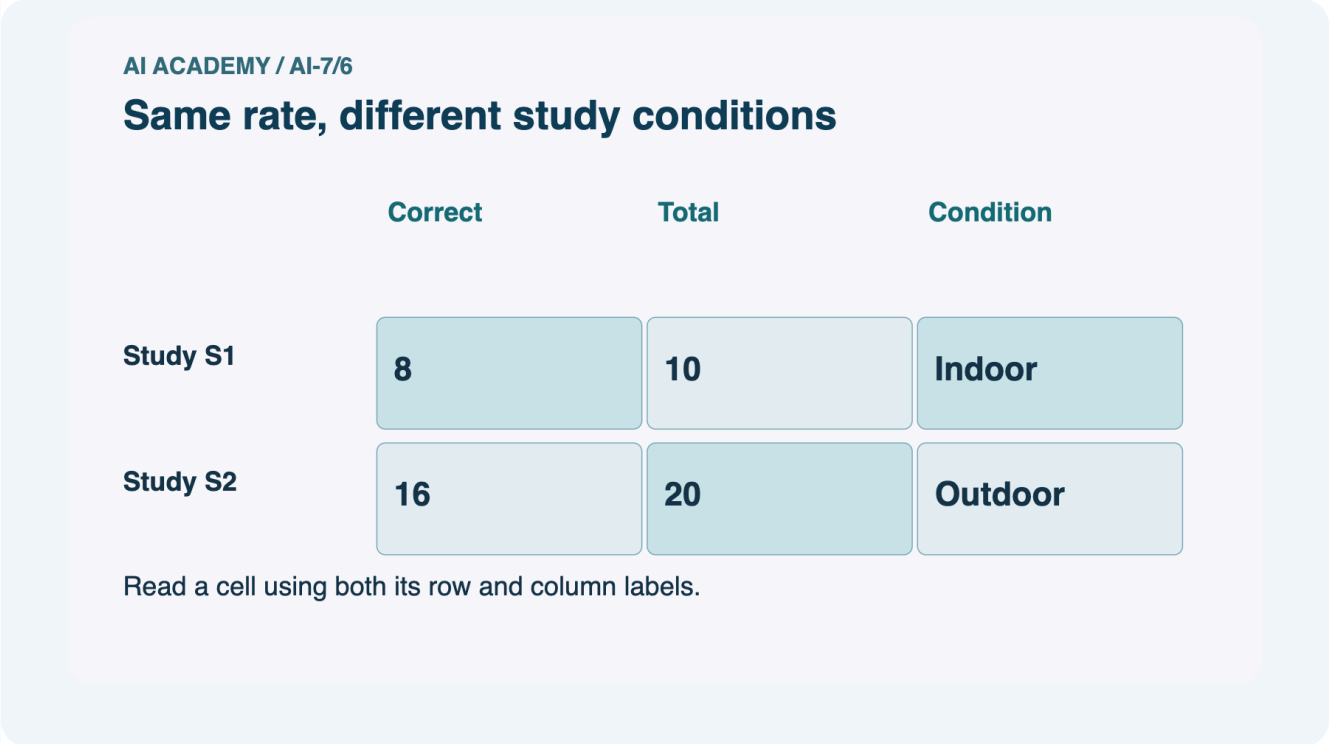
Follow the labelled picture with the matching explanation. Open the vector diagram

Record each method: Keep population, task, model version and scoring in separate columns.
→ Compare rates carefully: Both rates are **80%**, with different denominators and conditions.
→ Name missing comparison: Neither study directly compares both tools on the same photos.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Same rate, different study conditions



Follow the labelled picture with the matching explanation. Open the vector diagram

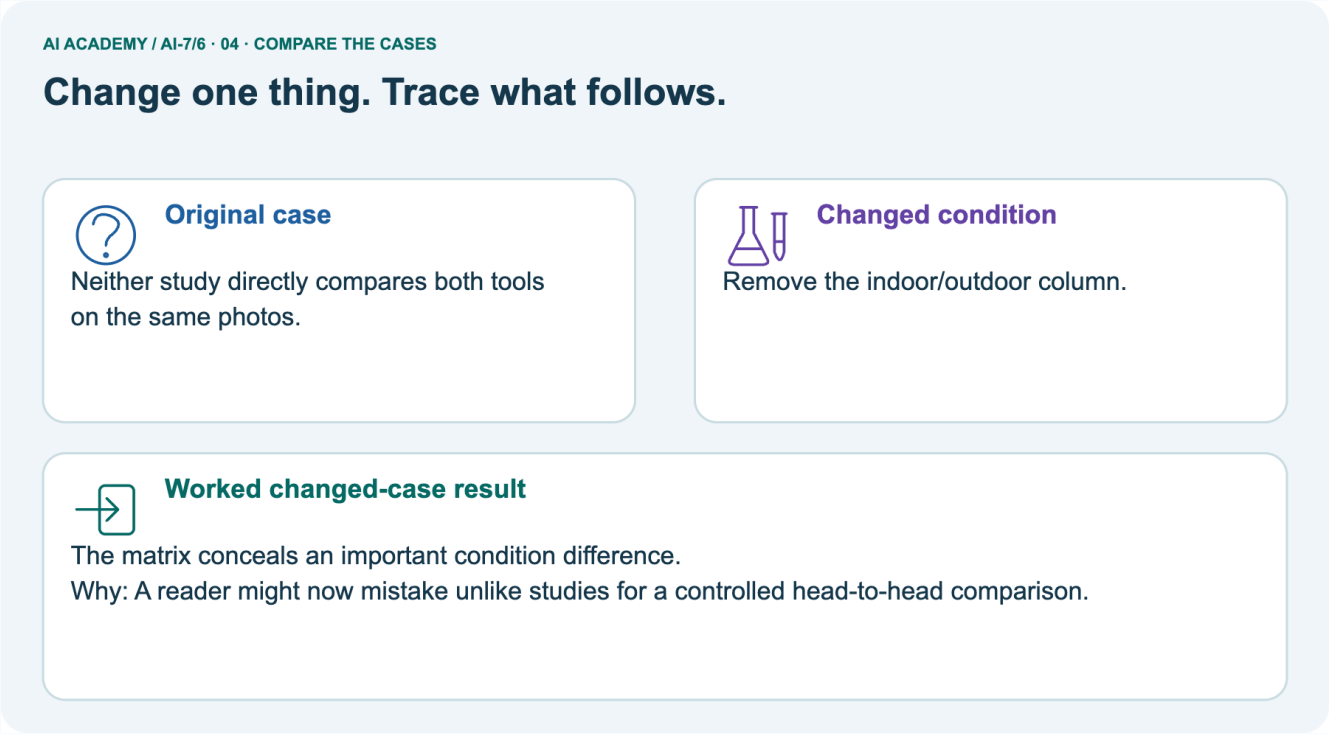
Same rate, different study conditions. Study S1: Correct = 8, Total = 10, Condition = Indoor. Study S2: Correct = 16, Total = 20, Condition = Outdoor.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Neither study directly compares both tools on the same photos. Changed condition: Remove the indoor/outdoor column. Expected result: The matrix conceals an important condition difference.

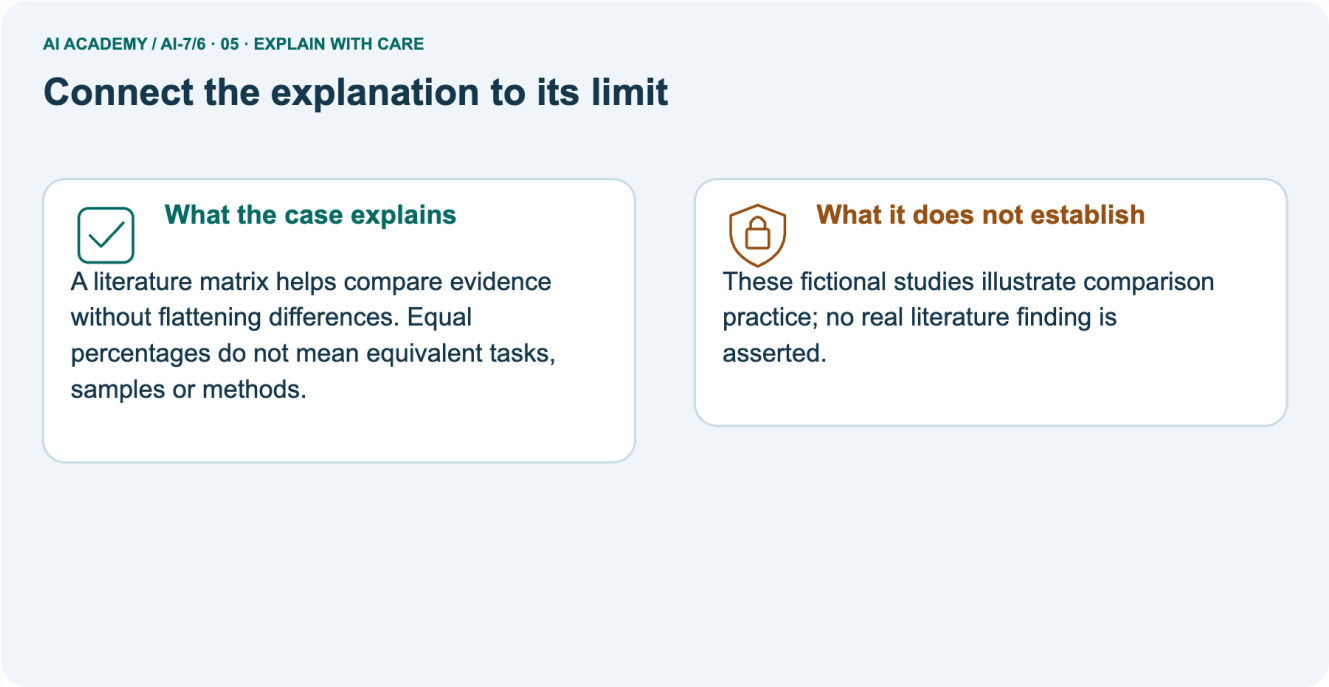
Reason: A reader might now mistake unlike studies for a controlled head-to-head comparison.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: A reader might now mistake unlike studies for a controlled head-to-head comparison.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **literature matrix** helps compare **evidence** without flattening differences. Equal percentages do not mean equivalent tasks, samples or methods.

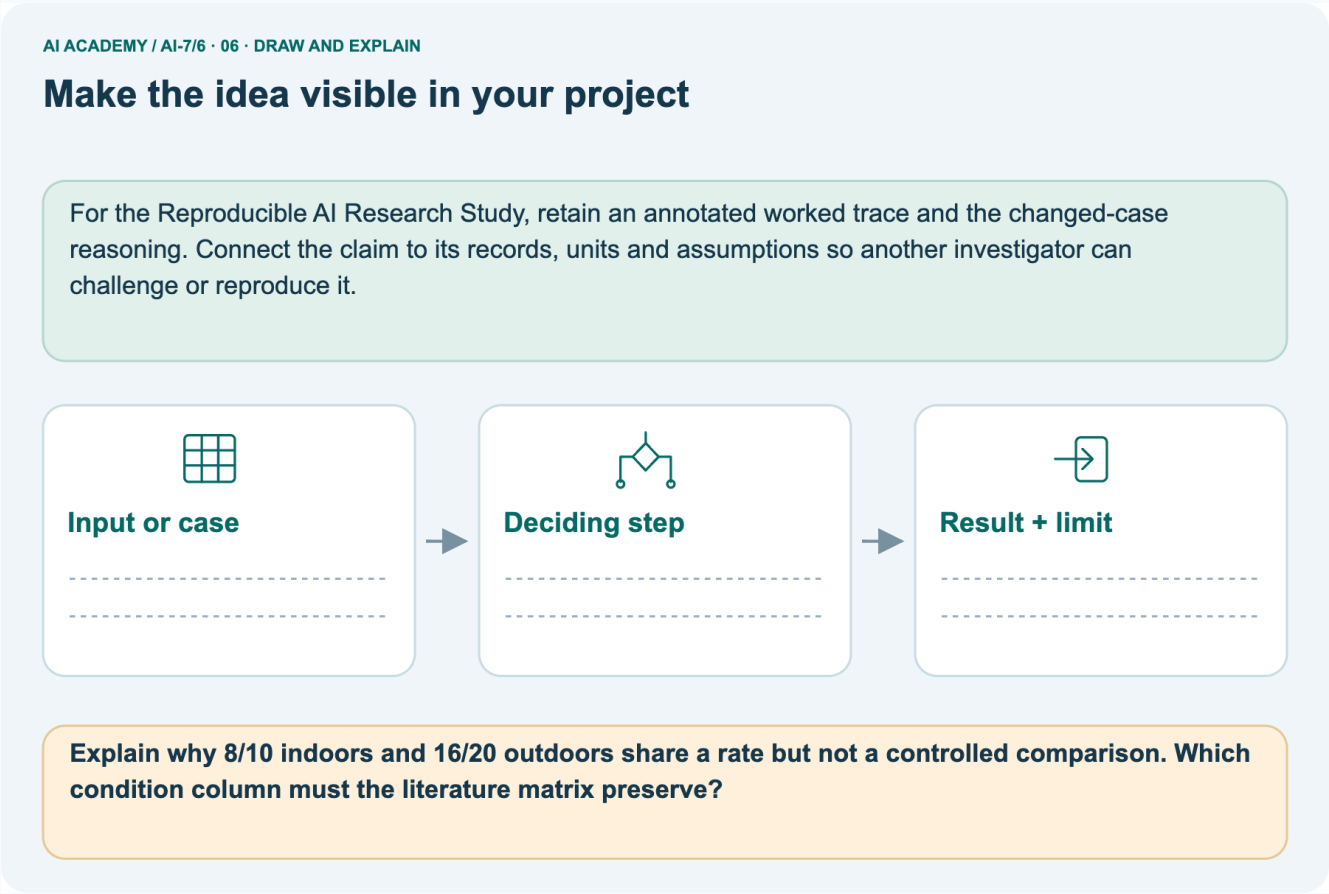
Limit: These fictional studies illustrate comparison practice; no real literature finding is asserted.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These fictional studies illustrate comparison practice; no real literature finding is asserted.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the **literature matrix** preserve?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Keep population, task, model version and scoring in separate columns. Both rates are **80%**, with different denominators and conditions. Neither study directly compares both tools on the same photos.

Changed case: The matrix conceals an important condition difference. A reader might now mistake unlike studies for a controlled head-to-head comparison.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. A literature comparison matrix: the case

02. Follow the deciding steps

03. Same rate, different study conditions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.